

Topic 01: How Machines "See" Data: Vectors & Norms

Applied Linear Algebra via NumPy

TOPIC 01

How Machines "See" Data: Vectors & Norms

High-dimensional feature space, dot products as projections, and L1 vs L2 geometry.

Cognitive Objectives

By the end of this topic, learners will be able to:

- **Represent multi-modal entities** (text embeddings, customer profiles, tensors) as directional arrows in $\mathbb{R}^d$.
- **Deconstruct dot products** visually as orthogonal shadow projections and angular similarity operators.
- **Derive geometric sparsity** by contrasting L_1 (diamond / Lasso) vs L_2 (hypersphere / Ridge) constraint topologies.
- **Vectorize distance computations** in NumPy avoiding Python loops for high-throughput batch similarity.

The High-Dimensional Similarity Dilemma

The Magnitude Trap: Euclidean Distance

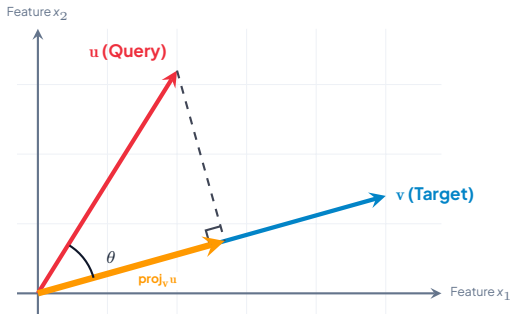
- Two users purchase identical products in identical ratios.
- User A: Wholesale distributor (volume = 10,000 units).
- User B: Individual consumer (volume = 2 units).
- Euclidean distance $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_2$ is **huge!**
- **Failure:** Euclidean distance confuses **scale (volume)** with **preference (direction)**.

The Geometric Discovery: Directional Alignment

- Isolate the angle θ between vectors in $\mathbb{R}^d$.
- Vectors pointing in the same direction have $\cos \theta = 1.0$, regardless of their individual lengths.
- Normalizing vectors projects them onto the unit hypersphere $\mathbb{S}^{d-1}$.
- **Solution:** Cosine similarity captures pure behavioral preference invariant to volume.

PUNCHLINE Normalize feature vectors when magnitude is uninformative.

Hero Visualization: Dot Product as Shadow Projection

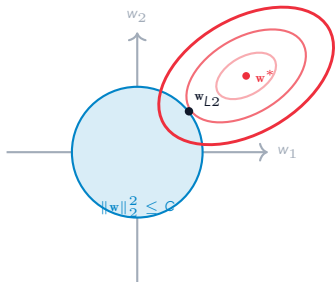


The Geometric Duality

- **Algebraic Definition:**
$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^d u_i v_i = \mathbf{u}^T \mathbf{v}.$$
- **Geometric Definition:**
$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \cos \theta.$$
- **Orthogonal Shadow:** The length of the shadow that $\mathbf{u}$ casts onto the span of $\mathbf{v}$ is:
$$\|\text{proj}_{\mathbf{v}} \mathbf{u}\| = \|\mathbf{u}\|_2 \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|_2}$$
- $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$ (Zero shadow / zero mutual information).

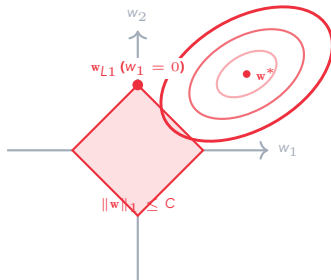
Hero Visualization: L_1 Diamond vs L_2 Circle Geometry

L_2 Ridge: Smooth Hypersphere



Mechanism: Tangency occurs along smooth curved boundary $\implies w_1 \neq 0, w_2 \neq 0$ (Smooth weight shrinkage).

L_1 Lasso: Diamond Sparsity



Mechanism: Loss ellipse hits the sharp diamond corner on the coordinate axis $\implies$ forces $w_1 = 0$ (**Exact Feature Selection**).

Production Checkpoint: High-Throughput Batch Similarity

Vectorized Similarity Matrix

Given M queries $A \in \mathbb{R}^{M \times d}$ and N items $B \in \mathbb{R}^{N \times d}$:

$$S_{M \times N} = \frac{AB^T}{\|A\|_2 \otimes \|B\|_2^T}$$

- Single matrix multiplication: $\mathcal{O}(MNd)$ ops.
- BLAS Level 3 cache-blocked execution (120× faster than nested loops).

PUNCHLINE Eliminate all Python loops
across feature rows.

Vectorized Recommendation Engine

```
1 import numpy as np
2
3 def batch_cosine_similarity(
4     A: np.ndarray, B: np.ndarray
5 ) -> np.ndarray:
6     """Pairwise cosine sim for M x d and N x d."""
7     # Compute dot products: shape (M, N)
8     dot_products = A @ B.T
9
10    # Compute L2 norms along feature axis
11    norm_A = np.linalg.norm(A, axis=1, keepdims=True)
12    # (M, 1)
13    norm_B = np.linalg.norm(B, axis=1, keepdims=True)
14    # (N, 1)
15
16    # Outer product broadcast division -> (M, N)
17    return dot_products / (norm_A @ norm_B.T)
```